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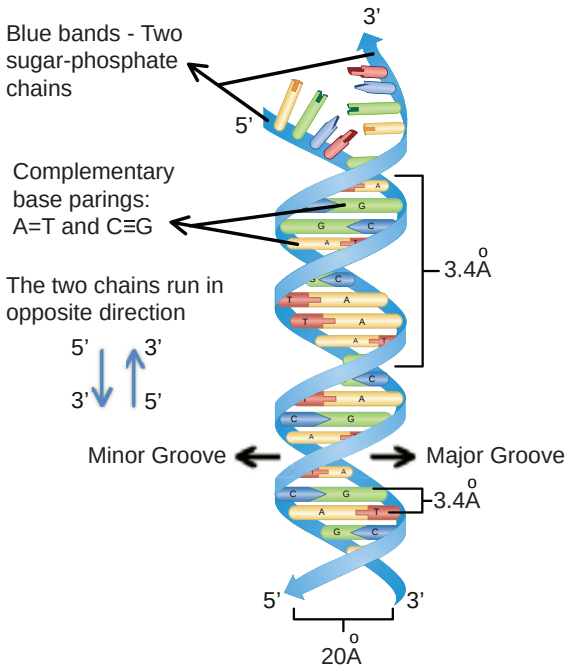



**Figure 8.43:** Structure of DNA

- Plectonemic coiling - the two strands of the DNA are wrapped around each other in a helix, making it impossible to simply move them apart without breaking the entire structure. Whereas in paranemic coiling the two strands simply lie alongside one another, making them easier to pull apart.
- Based on the helix and the distance between each turns, the DNA is of three forms – **A DNA, B DNA and Z DNA** (Figure 8.43) .

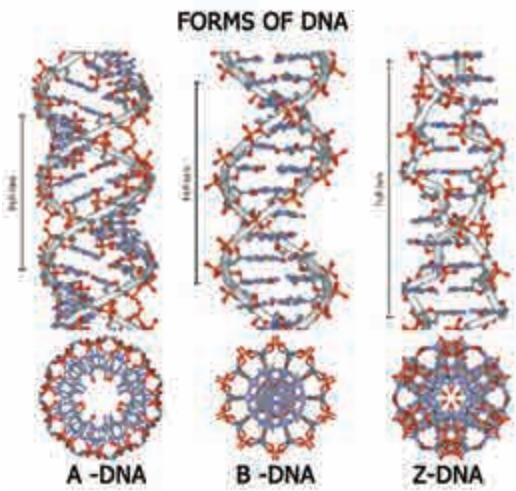



**Figure 8.44:** Forms of DNA

| Feature | B-DNA | A-DNA | Z-DNA |
|---|---|---|---|
| Type of helix | Right-handed | Right-handed | Left-handed |
| Helical diameter (nm) | 2.37 | 2.55 | 1.84 |
| Rise per base pair (nm) | 0.34 | 0.29 | 0.37 |
| Distance per complete turn (pitch) (nm) | 3.4 | 3.2 | 4.5 |
| Number of base pairs per complete turn | 10 | 11 | 12 |
| Topology of major groove | Wide, deep | Narrow, deep | Flat |
| Topology of minor groove | Narrow, shallow | Broad, shallow | Narrow, deep |

### 8.7.4 Ribonucleic Acid (RNA)

**Ribonucleic acid** (**RNA**) is a polymeric molecule essential in various biological roles in coding, decoding, regulation and expression of genes. RNA is single stranded and is unstable when compared to DNA (Figure 8.45).

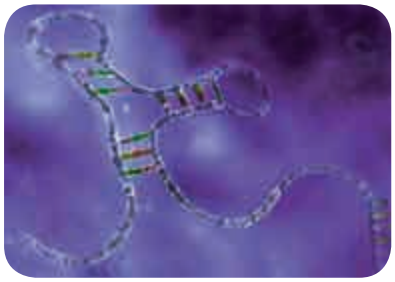


**Figure 8.45:** Structure of RNA